FIELD MANUAL

OPERATING NOTES // XX29 / FM-1

XENOLITH

One played note. A changing array of coupled oscillators.
Xenolith is a cell-based FM synthesizer for the M-VAVE FM-1. The instrument can grow from one to 64 paired oscillator cells, wire them in mixed topologies, alter the FM law and coupling medium, and route the result through a circuit-derived filter, VCA, folder and reverb.

01 / FIRST SOUND

Press a key. Release it to hear the envelope release. With SEQ disabled, PLAY latches a drone; PLAY again stops it. Begin quietly, then set the master volume. HOME leaves any page without restoring defaults.

The starting sound is 84 Hz, index 2.4, ratio 1.2, drift 0.20 and eight cells. The core runs at 44.1 kHz; the tank is 25% wet. Modulation, sequencing, filter, wavefolder and amplifier colour begin off.

```
[KEYS / MIDI / SEQ] -> [CORE ARRAY] -> [DIGITAL] -> [VCA / FOLDER]
                            ^                            |
[A / B / RANDOM / MIX] -----+                    [ECHO + TANK] -> OUT
[4x ADSR ENVELOPES] --------+      FILTER: PRE / POST / PRE FX / POST FX
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02 / PANEL MAP

HOME	Press: Main Panel	Hold: ARRAY (Press EDIT: COUPLING)
SELECT	Turn on HOME: Detail	Press on HOME: VCA -> WAVEFOLDER
EDIT	1x: DIGITAL 2x: FILT	Hold: Triode / Twin-T CIRCUIT
FX	1x: EXHAUST 2x: TANK	Hold: DIFFUSION | FX+REC: LOAD ENG
ENV	1x: 4x ADSR (Turn SEL)	Hold: Selected Envelope Routing
LFO	1x/2x: Main / Detail	Turn SEL: A/B/RAND/MIX (Hold: MEM)
SEQ	1x: Sequencer	Turn SEL: SYS/NOTE/CV (Press: Page)
ARP	1x: Key Assign / ANS	Press SEL: Range / Spectral Layout
GLO	1x: Spectrum (+EDIT)	Hold: Oscillator LOOKUP Table
PRESETS	Turn: Slots (PLAY=Ld)	SAVE 2x: Store | HOME+REC: Panic

03 / KEEP THE SOUND; LEAVE THE PAGE

Short HOME returns to the main panel while sound and control motion continue. REC records notes and parameter gestures, not an audio file. HOME + REC is the panic chord: it stops transport and clears the sound and tail, while keeping the take already recorded.

[READ IN ORDER, OR OPEN THE RELEVANT PLATE]

02 ARRAY + COUPLING / 03 KEYS + SEQ / 04 MOD / 05 FILTER + ENV
06 EXHAUST / 07 DIGITAL + TABLE / 08 VCA + FOLDER + LOAD + PATCHES

CORE ARRAY

CELLS / CONNECTIONS / COUPLING

01 / FOUR HOME CONTROLS & SELECT DETAIL

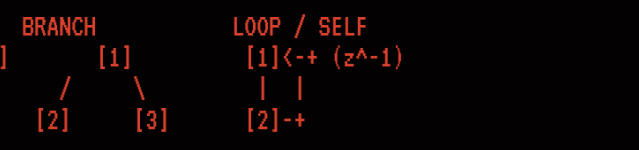
K1 TONIC 20-800 Hz. The central pitch of the array.
K2 FM INDEX 0-64 (0-120 st in EXP, 0-100% in WARP). Modulation depth.
K3 MOD RATIO 0.001-32. Modulator frequency divided by cell tonic.
K4 DRIFT 0-1. Independent slow pitch wander in the cells.

Rotate SELECT on HOME to open the four control detail pages: TONIC detail sets encoder response, sensitivity, GLIDE ms and PARAPHONY (MONO or 2-8 voices on eligible PARALLEL/SELF arrays). INDEX detail sets response, sensitivity, index and FM TYPE (or FLAVOUR). RATIO detail adds SNAP (OFF / 12-TET / RATIOS). DRIFT detail adds SPREAD X (0-2.0).

ALGORITHM adds or removes cells (1-64; up to 128 oscillators). More parallel cells first thicken the cluster, then open cubically across six octaves. Linked cells retain compact beating offsets.

02 / HOLD HOME: CURRENT VS GLOBAL ARRAY

K1 ADD/GROWTH PARALLEL / SERIES (CHAIN4) / BRANCH / LOOP (LOOP4) / SELF
K2 MODE LINEAR / PHASE / EXP / PULSE FM / PHASE WARP
K3 LINK Strength of forward inter-cell connection, 0-4.
K4 FEEDBACK Delayed z-1 return strength, 0-0.95, for LOOP or SELF.



CURRENT (ADD) remembers each cell's birth type. Rotate SELECT to choose AUTO or an earlier source cell C01-C64. Short SELECT previews a uniform GLOBAL (ALL) array; hold SELECT (650 ms) in GLOBAL to adopt it.

03 / PRESS EDIT ON ARRAY: FLAVOUR & COUPLING

K1 FLAVOUR NORMAL / RETURN - / RETURN +- / CASCADE (per FM mode).
K2 COUPLING OFF / BIAS / LAG / ALLPASS / COMMON bus.
K3 DEPTH Blend of the coupling transformation on the modulator.
K4 PARAMETER Bias drive, RC lag time, allpass delay or bus time.

RETURN - and RETURN +- invert feedback polarity for hollow odd harmonics. CASCADE attenuates direct bus output of upstream SERIES parents as LINK rises. COMMON degenerates every modulator against the delayed array mean.

[A USEFUL FIRST COMPARISON]

Build a 6-cell mixed array in CURRENT, then raise LINK and FEEDBACK. Press EDIT on ARRAY to compare NORMAL vs RETURN - and LAG vs COMMON.

KEYS + SEQUENCE

ASSIGN / MIDI / RECORD

01 / ARP IS THE KEYBOARD ASSIGNMENT PAGE

K1 DESTINATION Tonic, index, ratio, clocks, bits, filter and more.
K2 KEY MODE   GLOBAL scalar mapping or ANS CELLS band selection.
K3 LATCH   Keep a scalar assignment or band mask after key release.
K4 KEY GATE   Let key edges operate the main amplitude envelope.

Press SELECT for the second page. GLOBAL has LOW / HIGH / RESPONSE; LINEAR or LOG spreads that range across 27 keys. TONIC/CHROMATIC keeps ordinary note and octave behavior. Turning KEY GATE off lets keys reshape a latched PLAY drone without retriggering the envelope.

ANS CELLS selects bands within the array. Its second page sets LOW/HIGH spectral limits, BAND WIDTH and RELATIVE/SPECTRAL tuning. External USB and TRS MIDI notes play chromatically (including paraphony) and record into SEQ; MIDI Clock, Start, Continue and Stop sync transport.

02 / SEQ: 64-STEP NOTE TRACK AND FOUR CV LANES

Rotate SELECT: SYSTEM / TONIC NOTES / CV 1 / CV 2 / CV 3 / CV 4.
Press SELECT: alternate the selected object's main and detail pages.

SYSTEM: enable, tempo, length 1-64 (16 beats), division and QUANT. Its second page chooses record mode (STEP / REPLACE / OVERDUB), armed material and ADSR / CONTINUOUS note gate. CONTINUOUS supplies a smooth drone while transport runs, even through rests or CV-only patterns.

TONIC NOTES: select step, REST / NOTE / TIE, note and gate length. The second page can trigger function A, function B and RANDOM NEXT. CV 1 holds up to 64 steps; CV 2-4 hold 16 steps each with SLEW.

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STEP       01     02     03     04        64 steps / 16 beats
NOTES     [ C ]--[ G ]--[REST]--[ C ]      edit cursor != playhead
CV 1      [.2]---[.7]---[.4]----[.9] ----> INDEX, or any destination
CV 2      [  ]---[  ]---[  ]----[  ]       up to four control lanes
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03 / RECORDING A TAKE

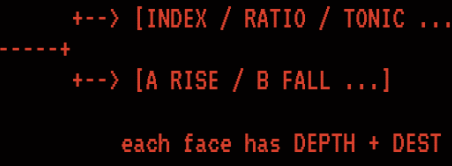
1. Stop transport for STEP entry, or use REPLACE/OVERDUB for live loops.
2. Press REC on HOME, SEQ or ARP to arm notes or the active CV target (if SEQ is off, REC arms a direct take that starts on the first key).
3. Play panel keys, MIDI notes, or turn a panel knob to capture values.
4. REC ends the take. PLAY auditions it. SAVE stores the complete patch.

MODULATION

TWO FUNCTIONS / RANDOM / ONE MIXER

01 / FOUR FACES, TWO PAGES EACH

Press LFO. Rotate SELECT to A, B, RANDOM or MIX. Short LFO alternates the selected face's pages. SELECT press bypasses the entire modulation system. A real destination creates its route and enables MOD; NONE removes that face's primary route. HOME leaves the patch running.



02 / RISE, FALL AND RANDOM MEMORY

A/B main: RISE / FALL / signed DEPTH / DESTINATION.
A/B detail: MODE / INPUT / TRIGGER / CURVE.
CYCLE repeats a rise and fall. AD makes a triggered contour. GATE follows a gate; SLEW follows an input with separate rising and falling times. OFF disables that function.

RANDOM main: RATE / SPREAD / DEPTH / DESTINATION.
RANDOM detail: CORRELATE / LEVELS / CLOCK / STEP-WALK output.
Hold LFO on RANDOM: LAW / MEMORY / RENEW % / BIAS.
CURRENT keeps the original law; BELL favors central values, EDGE the ends, BINARY two endpoints. MEMORY locks a loop of 2-32 events; RENEW 0% repeats the phrase, 100% replaces every visited event.

MIX main: INPUT A / INPUT B / DEPTH / DESTINATION.
MIX detail: GAIN A / GAIN B / OFFSET / OFF-SUM-PRODUCT-MAX.
The mixer combines control signals. It is not a second audio mixer.

03 / TWO SMALL PATCHES

MOVING INDEX: A in CYCLE, rise 0.05 s, fall 0.8 s, destination INDEX. Start with low depth. Send RANDOM WALK to FUNC A FALL, about +2 octaves, and choose FUNCTION END as its clock. Each ending chooses the next fall.

OPENING AMPLITUDE: leave A's own destination NONE. MIX input A=FUNC A, gain A=0.5, gain B=0, offset=-0.5, mode SUM; route MIX to AMPLITUDE, +1. PLAY with SEQ off supplies a drone. A now opens and closes its level.

[DEPTH UNITS ARE PART OF THE PATCH]

Pitch depth is semitones; ratio and time/rate depths are octaves. Index depth 1 spans 64 index units. Bits use signed bit offsets; CELLS adds or removes cells via slow prepared transactions.

CONTOUR + FILTER

FOUR ENVELOPES / ONE COMMON CIRCUIT

01 / ENV: ATTACK / DECAY / SUSTAIN / RELEASE

Rotate SELECT to AMP, FILTER, INDEX or TONIC. AMP is the shared amplitude envelope. The other three names are starting roles, not fixed wiring: hold ENV for DESTINATION / DEPTH / TRIGGER / ENABLE.

Choose KEY, SEQ NOTE or KEY + SEQ triggering. KEY includes the latched PLAY drone, which holds sustain until stopped. New contours begin disabled at zero depth. AMP's route is fixed; the other three are free.

02 / EDIT: DIGITAL / FILTER / HOME

FILTER: K1 CUTOFF / K2 MODE / K3 TOPOLOGY / K4 RESONANCE.
Rotate SELECT to OFF, PRE DIGITAL, POST DIGITAL, POST FX or PRE FX.
Press SELECT to cycle cutoff knob response (FINE / SWEEP / FAST / LINEAR).

```
CORE -> [PRE] -> QUANT/RECON -> [POST] -> VCA/FOLD -> [PRE FX] -> FX -> [POST FX]
        core clk                44.1k                 44.1k            stereo
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PRE DIGITAL filters before quantization. POST DIGITAL filters the quantized mono core. PRE FX filters mono after VCA/wavefolder colour and before the reverb split. POST FX shapes the stereo output and tail.

LP / BP / HP / NOTCH / PHASE use different outputs of a common reduced triode / parallel-T feedback model. TWIN T returns its selective plate response; CUTOFF becomes CENTRE. The curvature sits inside feedback, so level changes affect the response.

TOPOLOGY selects 2 POLE, 4 POLE or BRIDGE (ONE, TWO or BRIDGE in TWIN T). For TWIN T, try centre 200 Hz and resonance 0.7 on the 84 Hz drone.

03 / HOLD EDIT: CIRCUIT

K1 CATHODE R	470-10,000 ohm; default 2,200. Changes local degeneration in the loaded stage and its available feedback.
K2 HEADROOM	0.0625-1; default 0.25. Lower values bend at smaller input levels, with small-signal gain compensated.
K3 COUPLING	0.5-100 Hz; default 3.18. Raises the low-frequency coupling pole; higher settings remove more bass.
K4 T BALANCE	0.5-1; default 1. Moves linked passive component ratios; reducing balance fills the feedback network's notch.

These controls alter the shared cell beneath every filter response.

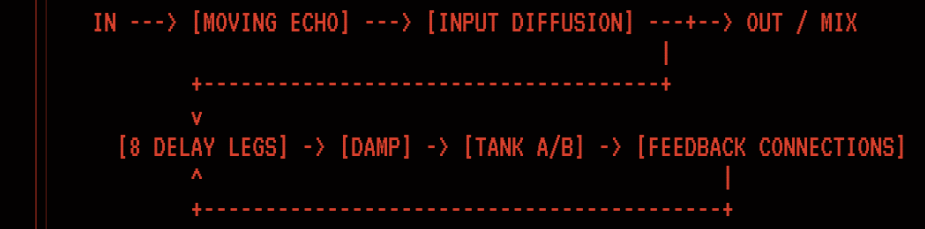
EXHAUST BUS

ECHO / FEEDBACK / DIFFUSION

01 / FX: THE FIRST TWO PAGES

K1 ECHO ms Moving delay before the tank. 2 ms to available memory.
K2 DAMP Hz High-frequency absorption in the recirculating paths.
K3 TAIL s Nominal low-frequency decay time of the feedback tank.
K4 MIX % Dry/wet balance.

Press FX again: WET kHz / STRUCTURE / DIFFUSION / WIDTH %.
Press SELECT on an FX page to change ZOH / LINEAR wet reconstruction.
Short FX from the second page returns HOME. Hold FX opens diffusion.



STRUCTURE changes how the stored sound is connected or heard. TANK is the base network. CLOUD4 / CLOUD8 expose additional early taps. RING circulates between legs; PAIRS couples them in pairs. The same delay memory continues through the change. WIDTH reduces the stereo spread toward mono; DIFFUSION softens the input before it enters the tank.

02 / HOLD FX: INSIDE THE TANK

K1 INPUT % The same input-diffusion control, 0-85%.
K2 TANK A % Short allpass sections in the first four feedback legs.
K3 TANK B % The corresponding sections in the other four legs.
K4 SCALE 1/4, 1/2 or 1 short-delay lengths.

A/B begin at zero. Raise one group first and listen to the tail after stopping a note. These sections redistribute delay across frequencies; they change the recirculating modes without clearing the long tail.

03 / THREE DIFFERENT WAYS TO MOVE TIME

ECHO sweeps move a read head. Its changing distance bends pitch and comb resonances, much like changing the speed of a recorded passage.

WET clock spans about 1.38-44.1 kHz. Lowering it changes sampling colour and stretches the fixed tank paths. ECHO can reach 1 s at 44.1 kHz, 2 s at 22.05 kHz, and up to 32 s at the slow end.

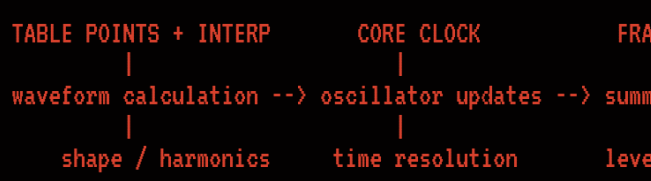
DIGITAL + TABLE

CLOCK / QUANTIZATION / WAVEFORM

01 / EDIT: THE DIGITAL PAGE

K1 MAX kHz Requested core rate: 1-44.1 kHz, then 88.2 / 176.4.
K2 FRAC BITS Native bypass at 24; turn down through the fractional grid.
K3 QUANTIZER TRUNC / ROUND / ERROR FB.
K4 RECON ZOH / LINEAR for lower-rate reconstruction (FIR at 2x/4x).

Press SELECT on DIGITAL to choose rate knob response (FINE / SWEEP / FAST / LINEAR). Output remains 44.1 kHz. Cell count and active modules limit the admitted rate; MAX is remembered and returns when work retires.



24 is native bypass. Values 2-23 quantize the summed signal on a fractional grid. At 1, the output becomes a polarity signal: -0.5 or +0.5, with exact zero retained. ERROR FB carries a bounded quantization error into later samples, making density part of the sound.

02 / HOLD GLO: OSCILLATOR TABLE CALCULATION

K1 TABLE POINTS 4-2048, powers of two; default 2048.
K2 INTERP ZOH / LINEAR; default LINEAR.
K3 HARMONIC 0-100%; blend toward a third-harmonic transformation.
K4 TARGET MOD / CAR / BOTH; choose which oscillator is altered.

Fewer points coarsen the waveform before it participates in FM. Four points with LINEAR form a triangle; ZOH makes a staircase. HARMONIC transforms a sine toward its third Chebyshev harmonic (3y - 4y^3).

03 / GLO: OBSERVE THE RESULT

The spectrum and phase display measures audio. PRESETS chooses L/R/MID; GLO then EDIT opens SPAN / TAP / FFT FPS / FFT SIZE. SELECT on HOME (when VCA/detail pages are closed) toggles the note-LED spectrum display.

COLOUR + LOAD

VCA / WAVEFOLDER / LOAD ENGINE / PATCHES

01 / SELECT FROM HOME: VCA AND WAVEFOLDER

Press SELECT on HOME to open VCA: K1 DRIVE % / K2 BIAS / K3 COLOUR % / K4 DEGEN % (local VCA degeneration). Rotate SELECT to choose the curve: CUBIC, SPACE3/2 (three-halves law), SOFTSIGN or TRI FOLD. COLOUR 0% bypasses the VCA curve exactly.

Press SELECT a second time for the independent diode WAVEFOLDER: K1 FOLD % / K2 BIAS / K3 STAGES (1, 2, 4, 6) / K4 MIX %. Press SELECT again to return HOME. FOLD is also a modulation/envelope destination.

02 / HOLD FX + REC: THE LOAD ENGINE

Holding FX and pressing REC opens the LOAD ENGINE (governor). It models physical voltage sag and progressively reduces processing demand when the 24 MHz cycle budget is stressed.

Press SELECT to toggle timing and detail views; rotate SELECT to set ONSET %. Hold REC (650 ms) to switch between effect-sag and core-relief pages (table bit loss, dither, interpolation, parallel voice aperture, link gain, filter sag, VCA bias and reverb shape). The REC lamp flickers organically when load relief is active.

03 / SIXTEEN PATCH SLOTS

PRESETS chooses 01-16 without immediately loading. PLAY on the patch page loads an occupied slot. SAVE, choose the slot, SAVE confirms. HOME cancels an unconfirmed save. Recall starts transport and REC stopped.

A patch stores sound, mixed array, coupling, flavours, key assignment, envelopes, modulation, sequence, load engine and effect settings.

04 / THE READING BEHIND THE CONTROLS

SMITH: sinusoidal FM, allpass dispersion and the DFT inform the core, coupling and spectral analysis.
WHITE / EARLY DSP: finite-wordlength arithmetic and error feedback inform the quantizer. Precision and clock are separate choices.
DATTORRO + SMITH: input versus tank diffusion inform the delay network.
RADIOTRON + SEELY: loaded triode stages, cathode degeneration, Twin-T networks and space-charge laws inform the filter, coupling and VCA.